

AKSENOVO-ZILOVSKOE, Russian Federation

WMO: 306690

Lat: 53.067N

Lon: 117.483E

Elev: 708

StdP: 93.10

Time Zone: 9.00 (E09)

Period: 86-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -34.9	(c) -32.3	(d) -37.7	(e) 0.1	(f) -34.6	(g) -35.2	(h) 0.1	(i) -31.8	(j) 5.7	(k) -18.2	(l) 4.8	(m) -21.0	(n) 0.8	(o) 240	(p) 0.645

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.9	27.2	16.8	25.4	16.2	23.8	16.1	19.1	23.9	18.2	23.0	17.3	21.8	2.1	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.5	13.7	21.0	16.6	12.8	20.1	15.5	12.0	19.4	57.8	23.9	54.6	23.1	51.7	21.8	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 6.3	(b) 5.3	(c) 4.4	(e) -36.9	(f) 30.9	(g) 3.3	(h) 1.5	(i) -39.2	(j) 31.9	(k) -41.1	(l) 32.8	(m) -43.0	(n) 33.6	(o) -45.3	(p) 34.7		
(a) 6.3	(b) 5.3	(c) 4.4	(e) -36.6	(f) 21.3	(g) 3.2	(h) 1.5	(i) -38.9	(j) 22.4	(k) -40.8	(l) 23.2	(m) -42.6	(n) 24.1	(o) -45.0	(p) 25.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.1	-22.8	-17.8	-10.5	-0.1	8.1	14.5	16.6	14.9	7.6	-1.3	-13.9	-21.0		
	DBStd	14.73	4.64	4.21	6.17	4.80	4.69	3.43	2.59	2.69	3.79	4.55	5.67	4.61		
	HDD10.0	4954	1016	779	636	303	93	6	1	1	91	350	717	962		
	HDD18.3	7461	1274	1013	894	552	319	119	63	111	321	608	967	1220		
	CDD10.0	556	0	0	0	1	33	142	207	153	20	0	0	0		
	CDD18.3	19	0	0	0	0	1	4	11	4	0	0	0	0		
	CDH23.3	452	0	0	0	3	51	151	171	74	1	0	0	0		
	CDH26.7	74	0	0	0	1	6	25	34	9	0	0	0	0		
	WSAvg	1.5	1.4	1.4	1.6	2.0	2.0	1.5	1.3	1.2	1.4	1.5	1.4	1.4		
	PrecAvg	434	6	5	6	21	39	73	102	90	56	17	13	7		
(15) Precipitation	PrecMax	597	16	16	11	53	75	132	225	144	100	45	31	21		
	PrecMin	327	1	1	0	6	5	28	31	55	14	1	2	3		
	PrecStd	62	3	3	3	13	17	29	42	24	24	9	6	4		
	PrecStd	62	3	3	3	13	17	29	42	24	24	9	6	4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-8.7	-3.8	10.6	20.1	27.3	29.1	30.1	28.0	23.1	15.3	5.2	-5.7		
		MCWB	-10.2	-6.3	3.8	9.3	14.7	18.2	18.1	18.7	13.9	7.8	1.1	-7.1		
	2%	DB	-11.7	-6.0	6.0	15.1	24.7	26.8	27.1	25.4	20.8	11.9	0.3	-9.7		
		MCWB	-12.7	-8.0	1.1	6.9	13.9	16.1	17.6	17.4	12.9	5.5	-2.3	-11.1		
	5%	DB	-13.8	-7.6	2.9	11.4	21.5	25.0	25.0	23.4	17.9	9.4	-2.7	-12.2		
		MCWB	-14.7	-9.3	-1.1	4.6	11.6	15.5	17.0	16.6	11.7	4.1	-4.7	-13.2		
	10%	DB	-15.7	-9.5	0.1	8.2	18.2	22.9	23.5	21.6	15.4	6.5	-5.3	-14.5		
		MCWB	-16.4	-10.9	-3.2	2.6	9.7	15.1	16.9	15.6	10.0	2.1	-7.0	-15.3		
	0.4%	WB	-10.1	-6.1	4.2	9.7	15.7	19.7	20.2	20.2	15.5	7.8	1.0	-7.0		
		MCDB	-8.7	-3.9	10.4	18.8	25.8	26.9	24.5	25.3	20.0	14.9	4.7	-6.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-12.7	-7.9	1.1	6.9	13.9	18.0	19.3	18.7	13.8	5.8	-1.9	-10.9		
		MCDB	-11.8	-6.2	5.8	14.8	23.5	24.1	23.8	23.3	18.0	11.1	0.0	-9.7		
	5%	WB	-14.6	-9.3	-1.1	4.8	12.0	17.0	18.6	17.5	12.7	4.2	-4.7	-13.2		
		MCDB	-13.8	-7.7	2.6	10.8	20.8	22.6	23.2	21.5	16.9	9.2	-2.7	-12.2		
	10%	WB	-16.4	-10.8	-3.0	3.0	10.3	15.9	17.7	16.4	10.8	2.3	-6.9	-15.2		
		MCDB	-15.7	-9.5	0.0	7.2	17.7	20.7	22.2	20.2	15.3	6.2	-5.4	-14.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.2	14.1	14.4	12.5	14.0	12.9	11.9	11.7	12.3	12.3	11.6	9.7		
		MCDBR	12.5	15.6	15.4	18.3	20.1	17.9	16.2	15.8	17.3	15.5	12.7	11.4		
		MCWBR	11.5	13.8	11.2	11.4	10.6	8.8	8.3	8.3	10.0	9.9	10.2	10.3		
	5% WB	MCDBR	12.3	15.5	14.7	17.2	19.2	13.8	12.6	12.7	14.5	15.2	12.8	11.5		
		MCWBR	11.5	13.7	10.8	10.8	10.3	7.8	7.4	7.4	8.7	9.9	10.3	10.3		
(40) Clear-Sky Solar Irradiance	taub		0.246	0.274	0.318	0.423	0.532	0.484	0.428	0.397	0.352	0.307	0.272	0.252		
	taud		2.194	2.167	2.131	1.989	1.776	2.013	2.244	2.309	2.377	2.394	2.255	2.133		
	Ebn at Noon		743	830	858	799	725	774	818	823	822	789	699	643		
	Edh at Noon		64	91	118	157	206	164	128	114	94	74	60	56		
	RadAvg		1.31	2.46	4.16	5.23	5.56	5.96	5.44	4.68	3.61	2.44	1.41	0.99		
(45) All-Sky Solar Radiation	RadStd		0.03	0.04	0.12	0.31	0.40	0.42	0.41	0.27	0.27	0.17	0.07	0.03		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page